

Computer Communication Networks

Session 1

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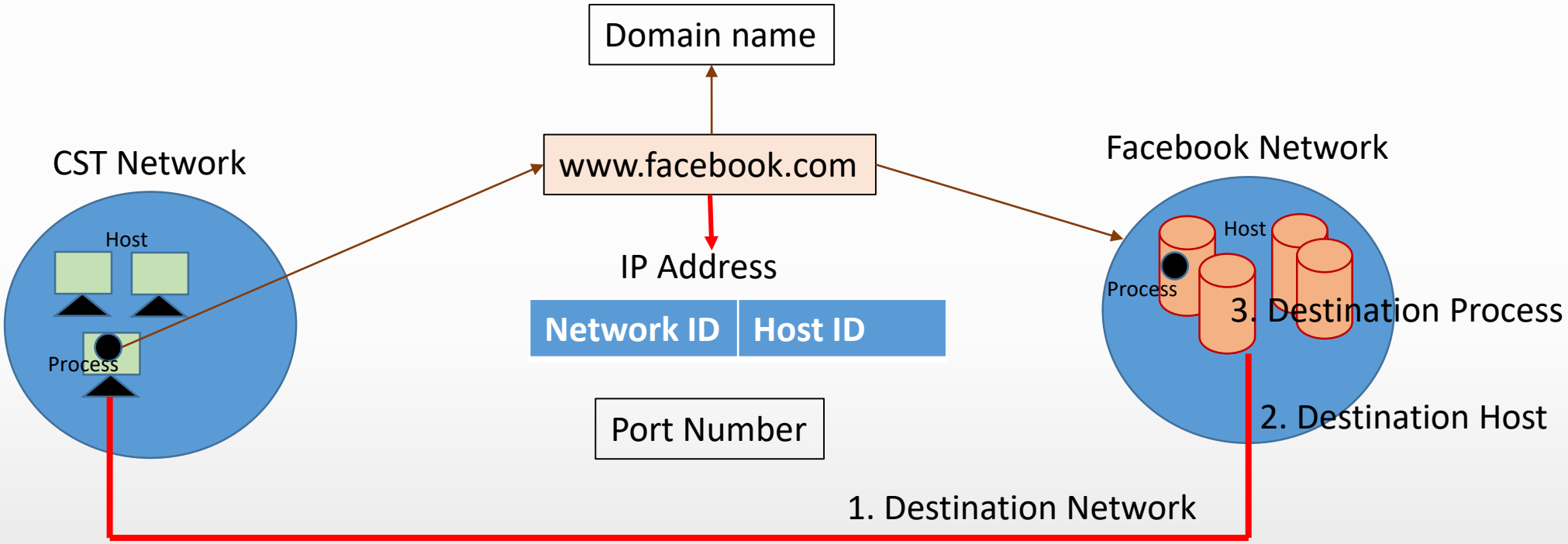


Introduction to Computer Networks

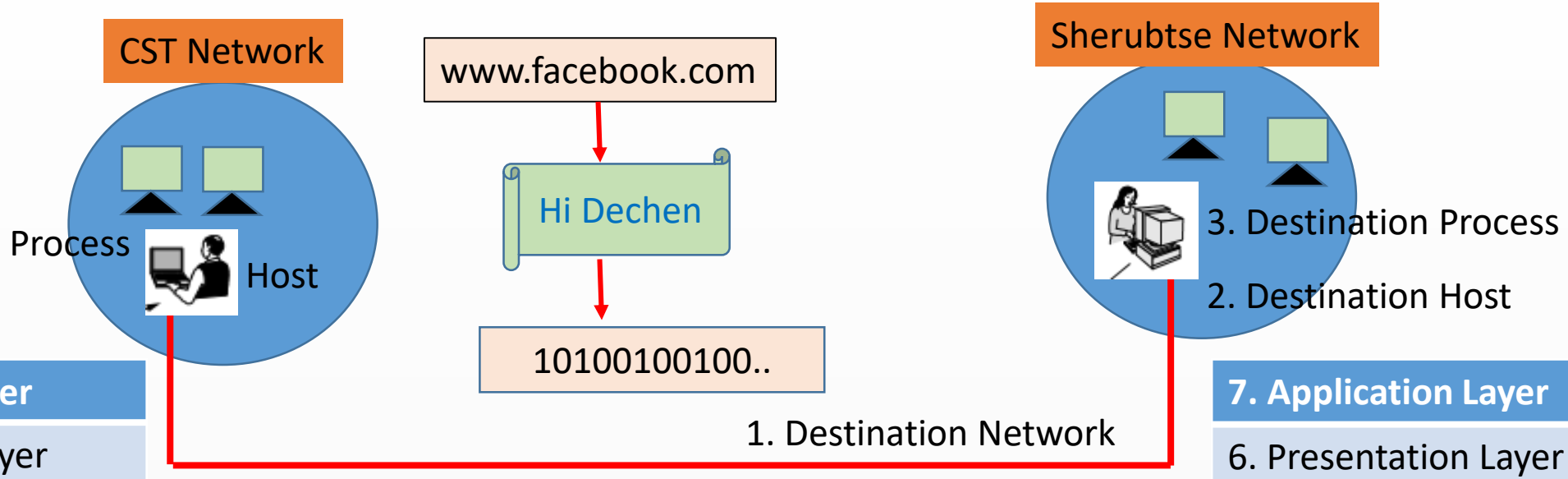
Learning Outcome:- At the end of the lesson, you are expected to be able to:

1. Overview
2. Define Computer Network
3. Identify Goals of Networking
4. Appraise Applications of Networking
5. Compare Network Topologies
6. Demonstrate Transmission Modes
7. Computer Applications
8. Network Types
9. Internetworks

Overview:-How does Computer Network Works?



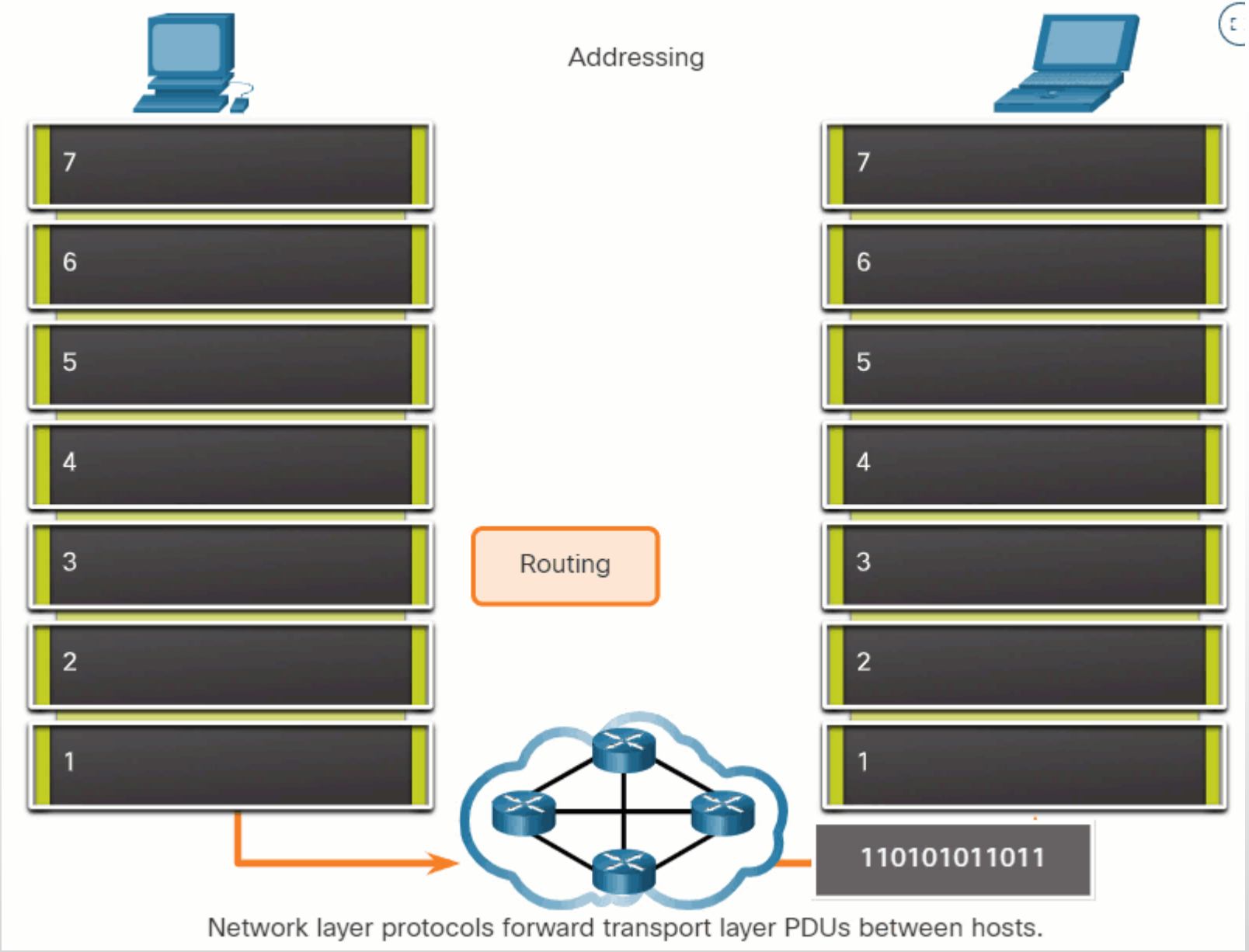
Overview:-How does Computer Network Works?



7. Application Layer
6. Presentation Layer
5. Session Layer
4. Transport Layer
3. Network Layer
2. Data Link Layer
1. Physical Layer

7. Application Layer
6. Presentation Layer
5. Session Layer
4. Transport Layer
3. Network Layer
2. Data Link Layer
1. Physical Layer

Overview:-How does Computer Network Works?



Overview:- OSI Layer

1. **Physical Layer**: This layer is responsible for sending and receiving bits of data between devices.
2. **Data Link Layer**: This layer makes sure that data is sent correctly from one device to another.
3. **Network Layer**: This layer is responsible for routing data packets between networks.
4. **Transport Layer**: This layer makes sure that data is sent correctly from one application to another.
5. **Session Layer**: This layer sets up, manages, and ends connections between applications.
6. **Presentation Layer**: This layer makes sure that data is in the right format for the application to understand.
7. **Application Layer**: This layer is where the user interacts with the network.

Overview:- Writing email to your friends

1. **Application Layer**: You open your email application and compose a new message to your friend.
2. **Presentation Layer**: The email application formats the message in a way that can be understood by the network.
3. **Session Layer**: The email application establishes a connection with the email server.
4. **Transport Layer**: The email message is split into smaller packets of data and sent to the email server.
5. **Network Layer**: The packets are routed through the network to reach the email server.
6. **Data Link Layer**: The packets are sent correctly from one device to another until they reach the email server.
7. **Physical Layer**: The bits of data are sent over the physical connection between devices.



1. What is Computer Networks?

- Interconnections of computers and other hardware components by transmission media either:

    1. Guided Transmission Media.
    2. Unguided Transmission Media.

used for sharing resources and information.



3. Networks Goal

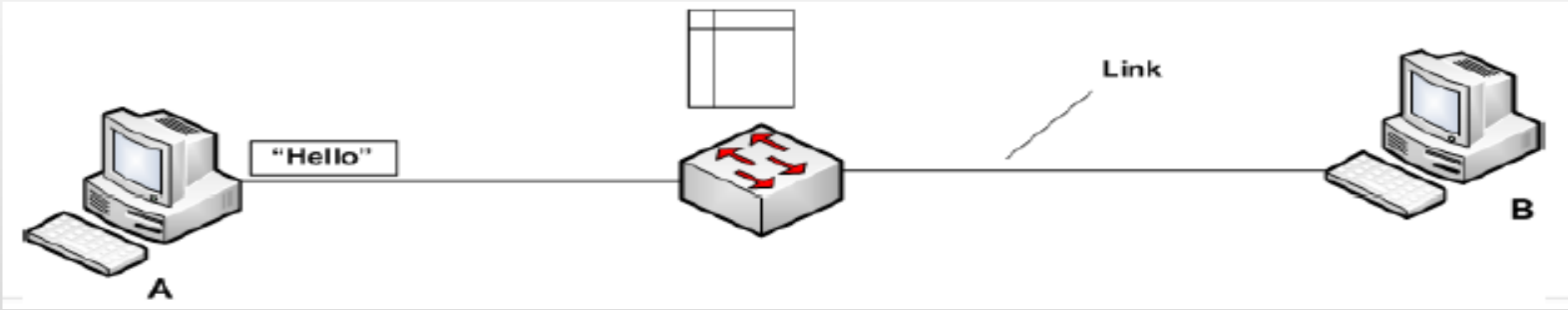
- Goals of the networks:

1. Resource Sharing- main goal
2. High Reliability
3. Cost Reduction
4. Improve Performance
5. Communication:
   - Data Communication



Data Communication Components

- Five Components:
  1. Message: Information(data) to be communicated.
  2. Sender: Device that sends the data message.
  3. Receiver: Device that receives the data message.
  4. Medium: Transmission medium, the physical path by which a message travels from sender to receiver.
  5. Protocol: Set of rules that government data communication





4. Applications of Computer Networks

Some of the Applications:

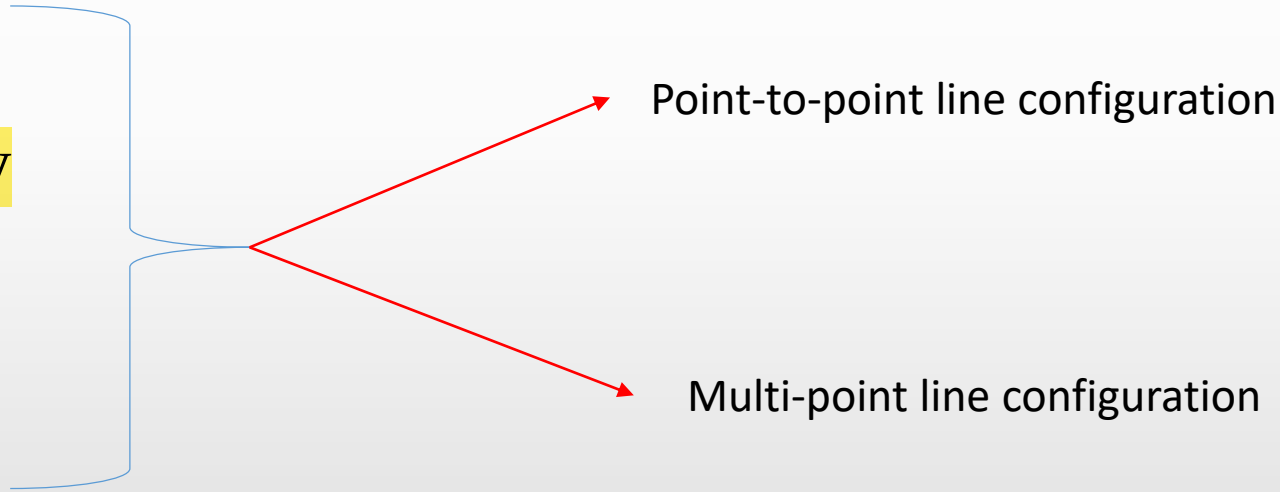
- Marketing & Sales
- Financial services
- Manufacturing
- Information Services
- Electronics Data Transfer
- Teleconferencing
- VoIP, Social networking, etc.



5. Network Topology

- The way in which a network is laid out physically.
- The four basic topologies are:

1. Star Topology
2. Mesh Topology
3. Bus Topology
4. Ring Topology





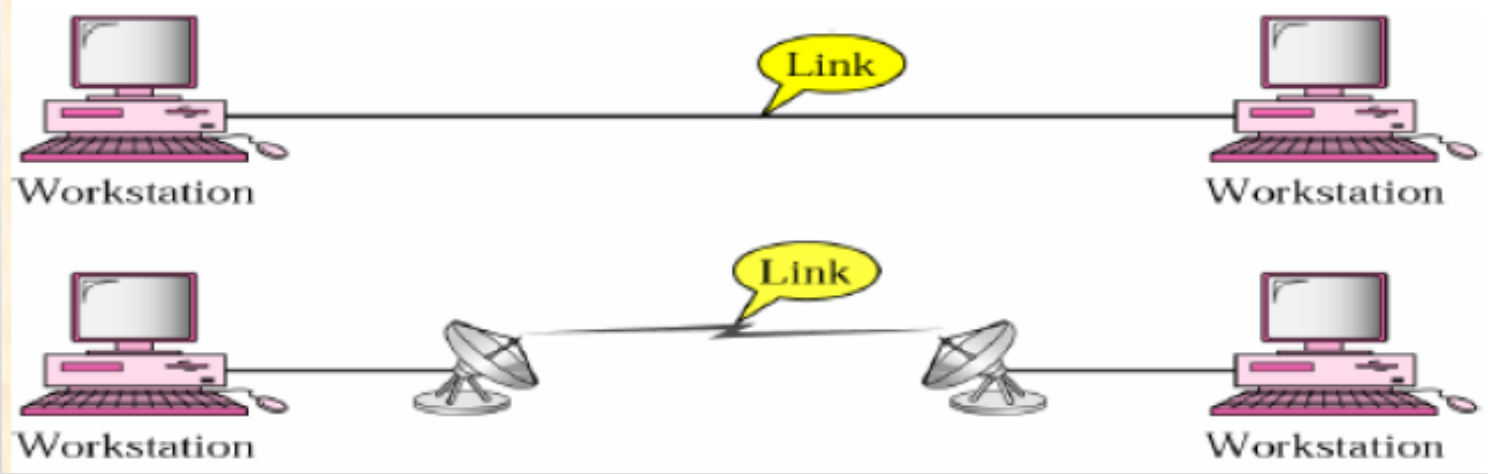
Network Topologies

❖Two types of line configuration are:

1. **Point-to-point** (Mesh , Star and Ring)
2. **Multipoint** (Bus)

➢ Point-to-Point Line Configuration

Point-to-point connections provide a dedicated link between two devices. Each device can communicate only with those that are directly connected to it.

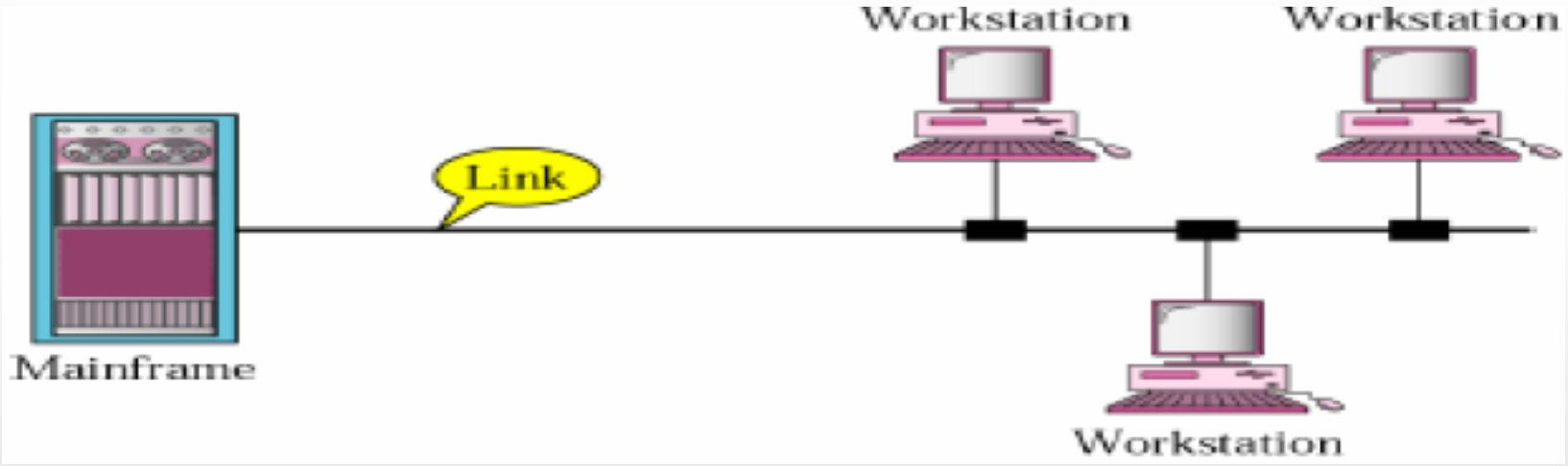




Network Topologies

Multipoint Line Configuration

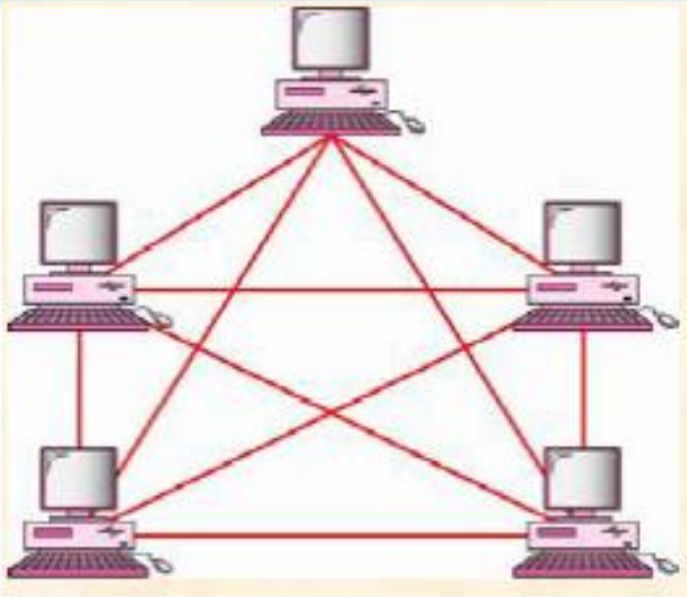
In multipoint networks many devices share a single link or communication medium.





Network Topologies

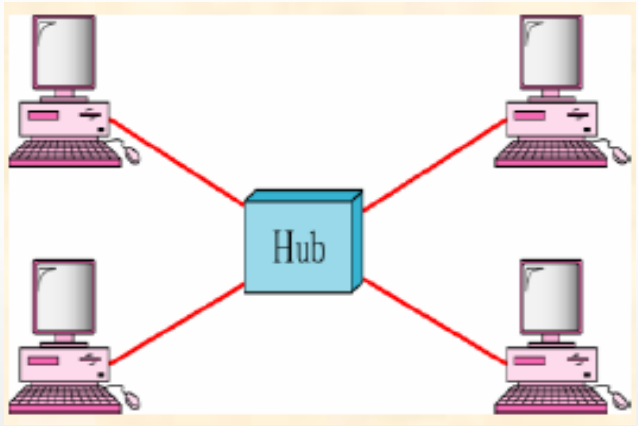
- **Mesh Topology**
  - If one connection fails, other connections will still work.
  - No. of cable=n(n-1)/2.
  - No. of ports= n-1





Network Topologies

- **Star Topology**
  - The other devices will still be able to send data through the hub; there will be no access to the device which has the failed connection to the hub.



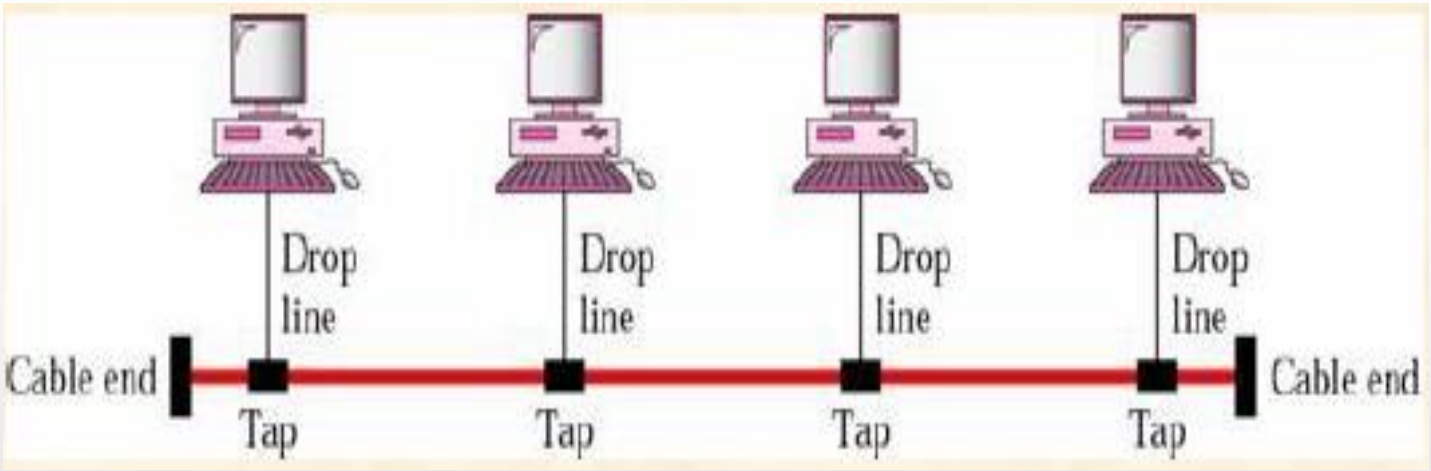
  - No. of cable= n
  - No. of port= n



Network Topologies

Bus Topology

- All transmission stops if the failure is in the bus. If the drop-line fails, only the corresponding device cannot operate.



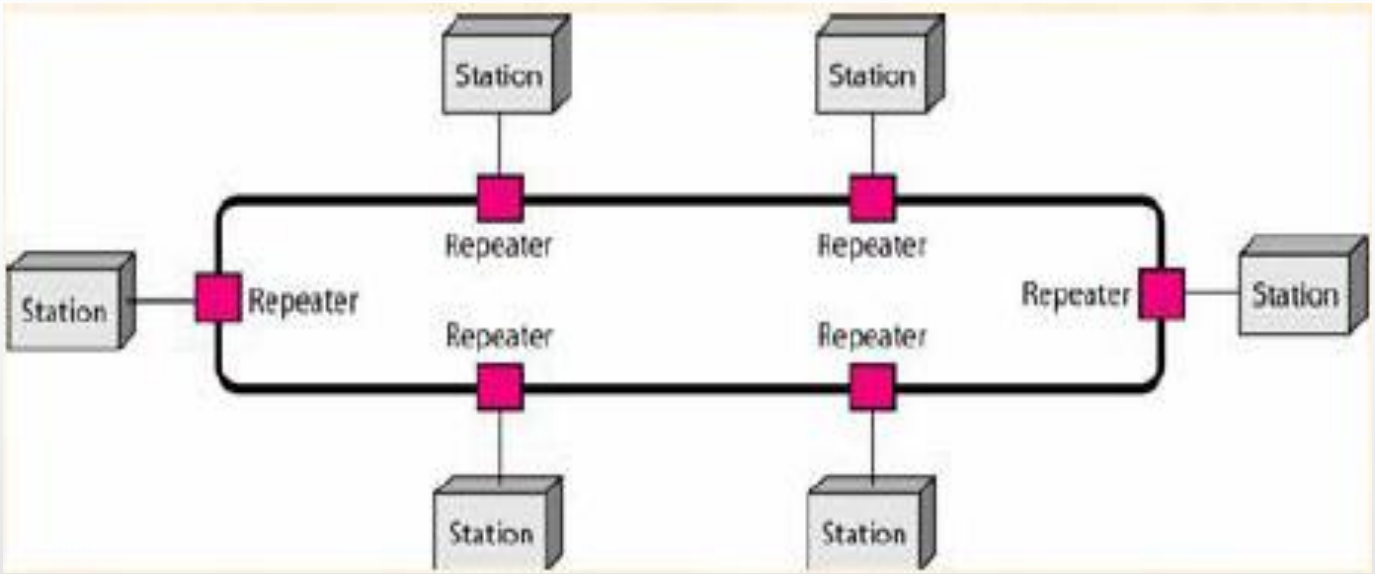
- No. of cable= One backbone and n drop lines



Network Topologies

Ring Topology

- The failed connection may disable the whole network unless it is a dual ring or there is a by-pass mechanism.



- No. of cable= n



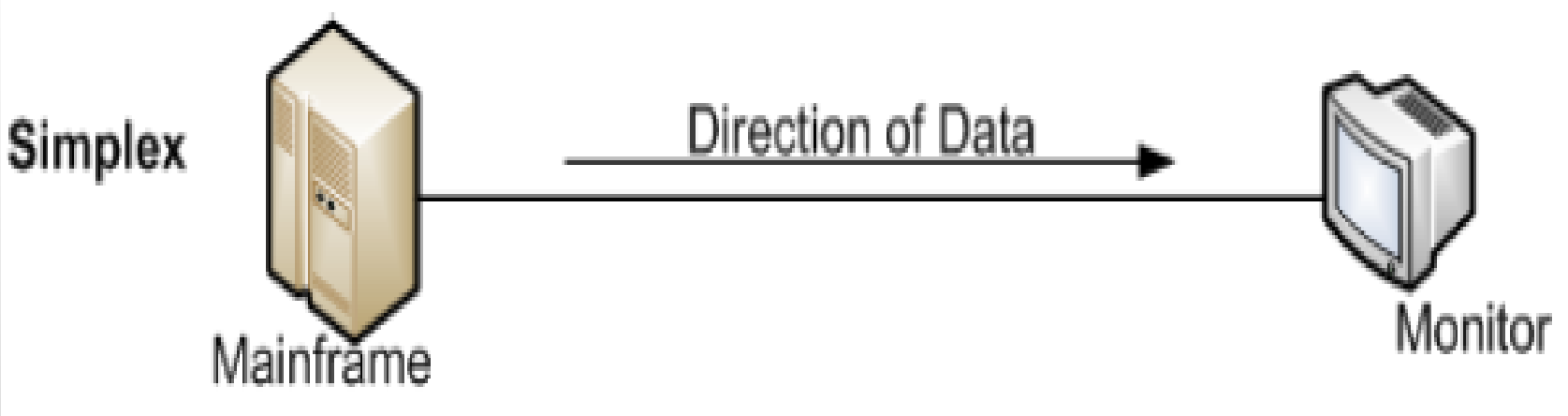
6. Transmission Mode

- Transmission Modes: It is also called as Communication Mode
  - Transferring of data between two device.
  - These modes direct the direction of flow of Information
  - It is used to define the direction of signal flow between two linked devices.
- Three types of transmission modes:
  1. Simplex
  2. Half-Duplex
  3. Full-Duplex



Transmission Mode: Simplex

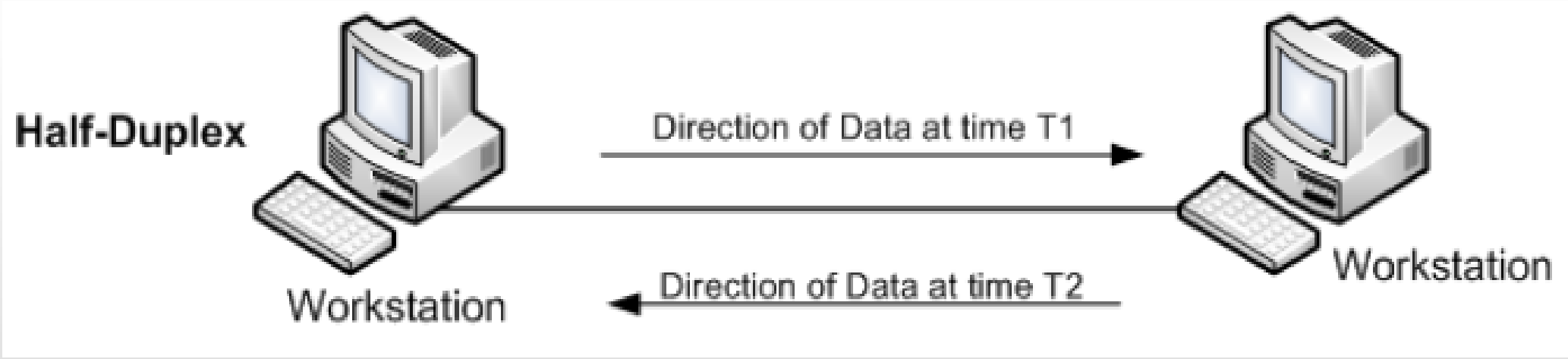
- Simplex:
  - Communication Channel that send information in one direction only.
  - Unidirectional, as on a one-way street.
  - One can transmit and other can only receive
  - Example: Key board and Monitor, Radio, TV, BBS





Transmission Mode: Half-Duplex

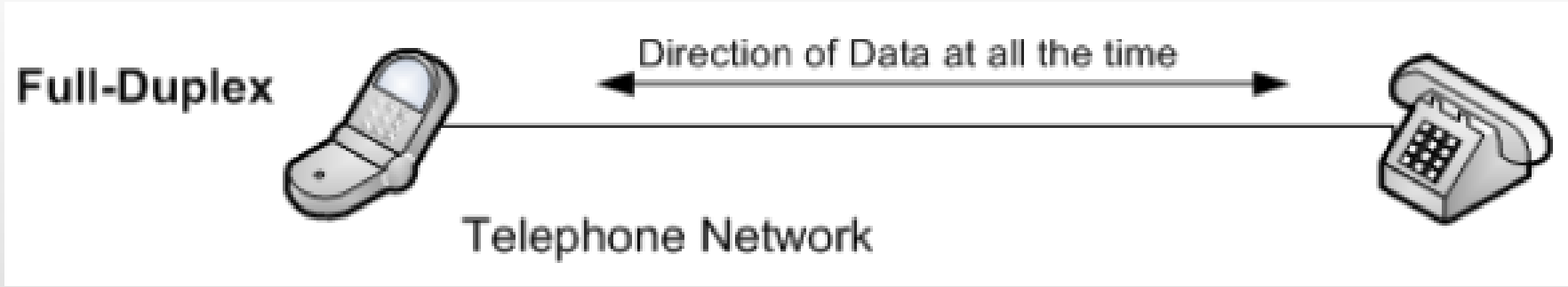
- Half-Duplex:
  - Bidirectional but not at the same time.
  - Each station can both transmit and receive & vice-versa.
  - Example: Walkie-Talkie





Transmission Mode: Full-Duplex

- Full-Duplex:
  - Bidirectional, both stations can transmit and receive simultaneously.
  - Example: Telephone Network, Mobile Conversation



Computer Network Applications

1. Business
2. Home
3. Mobile Users
4. Social Users

Computer networks play a crucial role in modern businesses, enhancing efficiency and enabling new opportunities

- Enterprise Resource Planning (ERP)
- Customer Relationship Management
- Supply Chain Management
- Telecommuting and Remote Work
- File Sharing and Collaboration

Computer Network Applications

1. Business
2. Home
3. Mobile Users
4. Social Users

Home networks have become essential for everyday activities, providing connectivity for various devices and services.

- Internet Access
- Home Automation
- File Sharing and Printing
- Streaming Services
- Online Gaming

Computer Network Applications

1. Business
2. Home
3. Mobile Users
4. Social Users

Mobile networks cater to users on the go, offering connectivity and services through smartphones, tablets, and other portable devices

- Email and Messaging
- GPS and Navigation
- Mobile Apps
- Social Media
- Mobile Payment

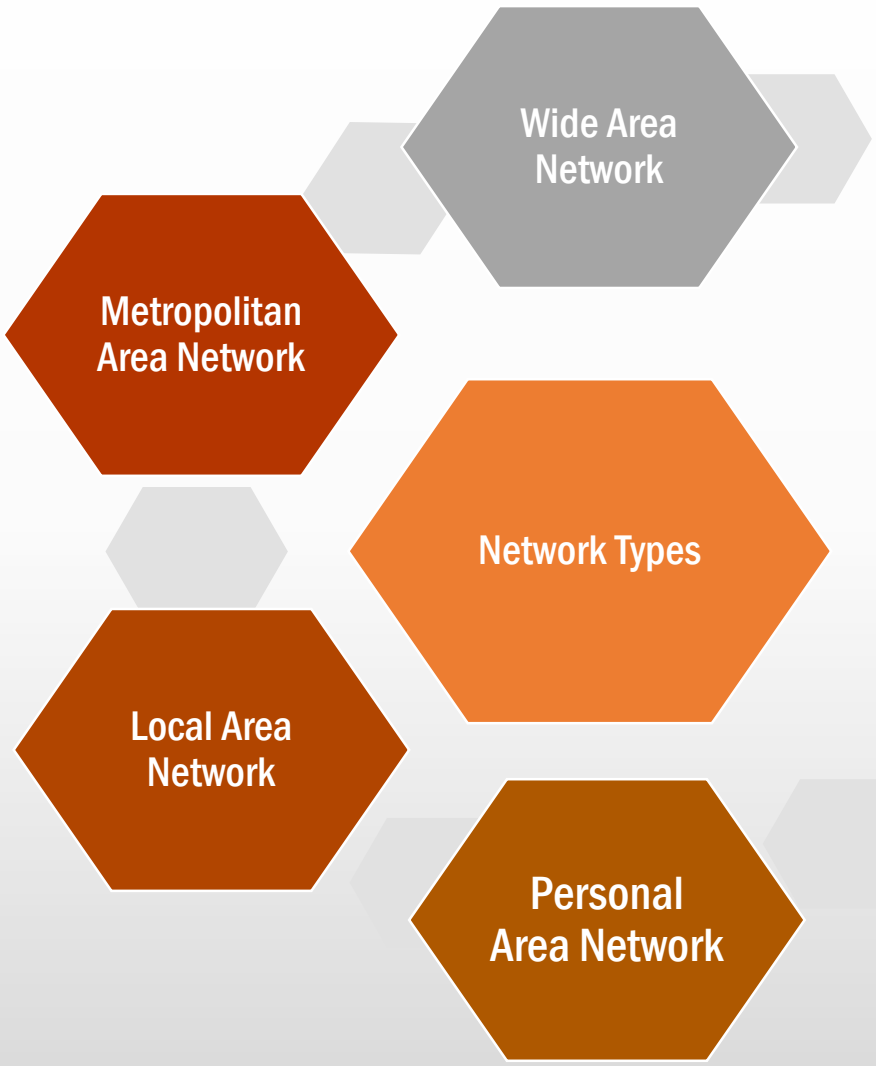
Computer Network Applications

1. Business
2. Home
3. Mobile Users
4. Social Users

Social networks and online communities facilitate interaction, content sharing, and collaboration among users.

- Social Media Platforms
- Online Communities
- Content Sharing
- Gaming Communities

Computer Network Types



Personal Area Network

A network for interconnecting devices centered around an individual person's workspace.

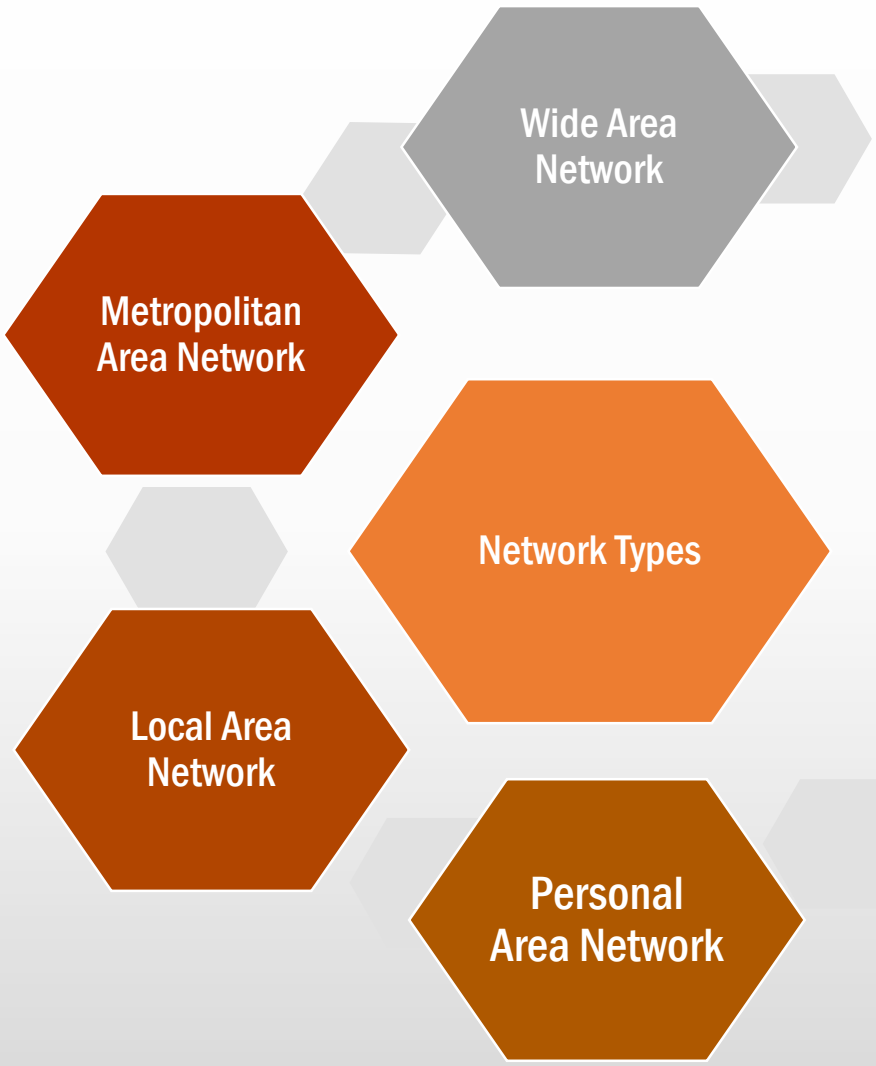
Range: Typically within a range of a few meters.

Examples: Bluetooth connections, USB connections.

Uses: Connecting devices like smartphones, tablets, laptops, and wearable devices.

Benefits: Convenience, ease of connection, low power consumption.

Computer Network Types



Local Area Network

A network that connects devices in a limited geographic area, such as a building or campus.

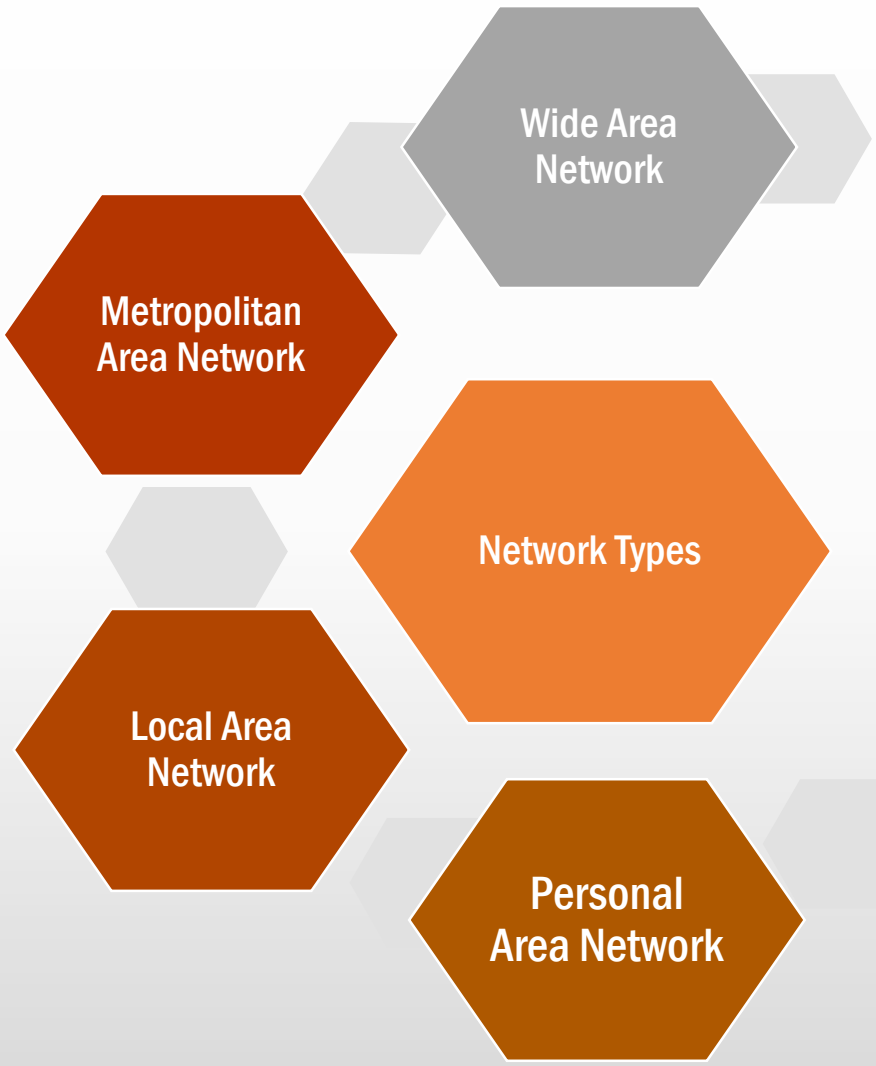
Range: Typically spans up to a few kilometers.

Examples: Ethernet networks in offices, Wi-Fi networks in homes.

Uses: Connecting computers, printers, servers within a localized area.

Benefits: High speed, low latency, cost-effective.

Computer Network Types



Metropolitan Area Network

A network that spans a city or a large campus, connecting multiple LANs.

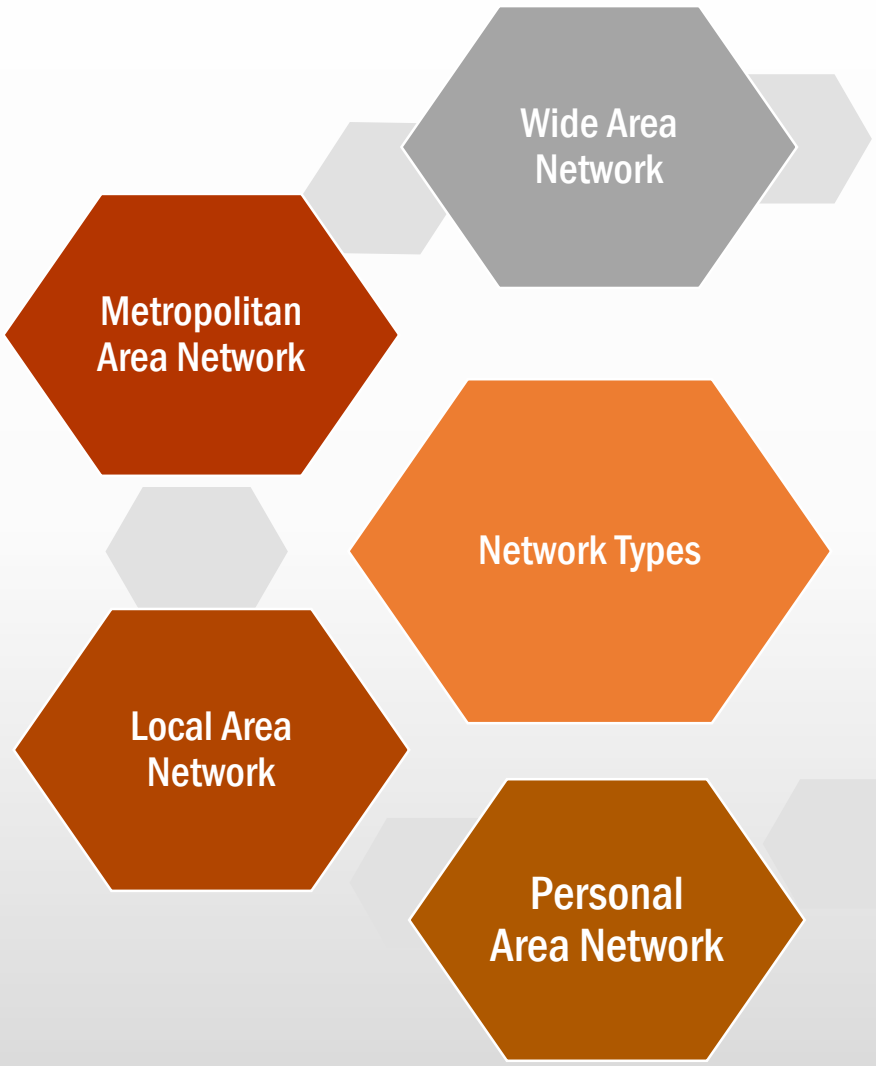
Range: Typically spans up to 50 kilometers.

Examples: City-wide Wi-Fi networks, university campus networks.

Uses: Providing network connectivity to a larger geographic area than a LAN.

Benefits: High bandwidth, scalability, coverage over a larger area.

Computer Network Types



Wide Area Network

A network that covers a broad area, often a country or continent.

Range: Can span thousands of kilometers.

Examples: The Internet, corporate networks with global reach.

Uses: Connecting geographically dispersed offices, providing Internet access.

Benefits: Long-distance connectivity, extensive reach, enabling global communication.

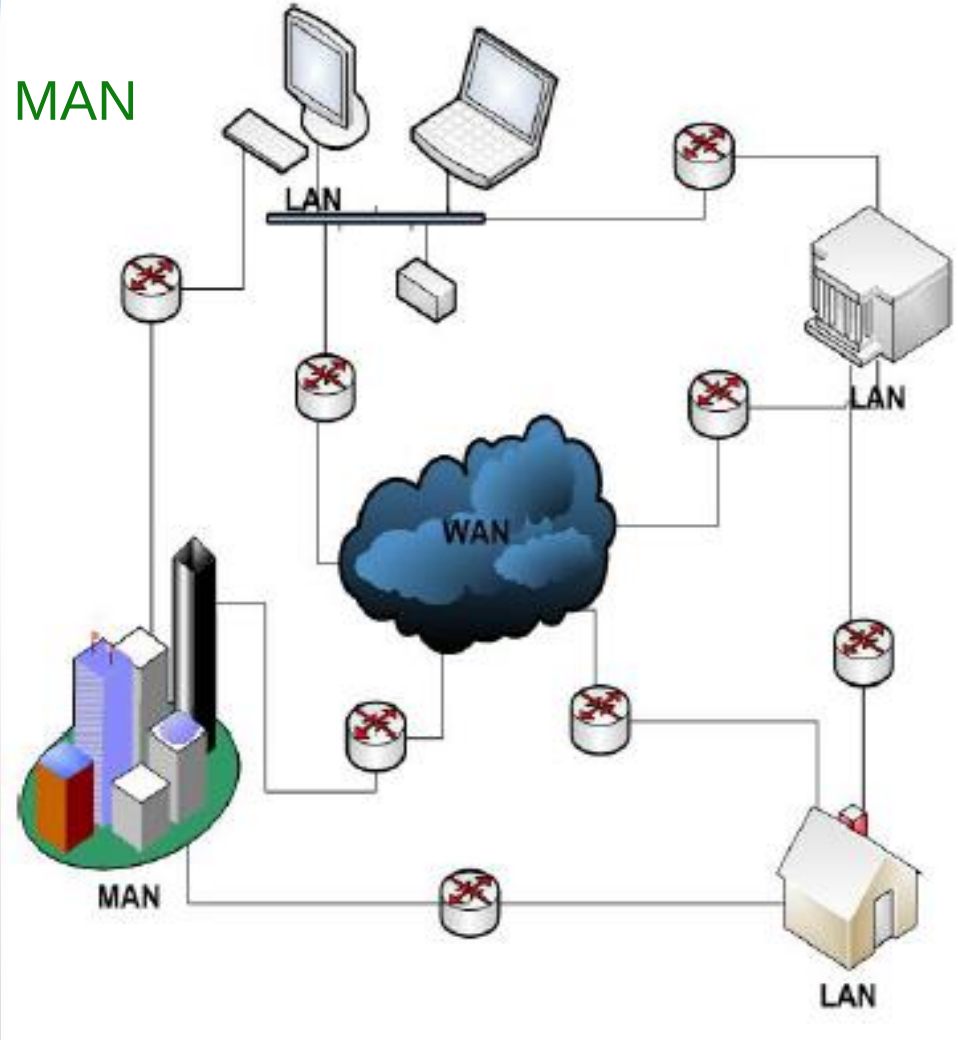


8. Internetwork

- Internetworks:

All LANS together makes MAN

  - Connection of two or more networks are known as internetworks or internet.
  - The term internet (lower case i) & Internet (upper case I) are different.
    - The internet is a generic term used to mean an interconnection of networks
    - Internet is the name of a specific world wide network.







References

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